# Solving Linear Equations

Exam code: 0607

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# Solving linear equations

## What are linear equations?

- A **linear equation** is one that can be written in the **form** $ax + b = c$
  - $a$, $b$, and $c$ are **numbers** and $x$ is the **variable**
    - $2x + 3 = 5$
    - $3x + 4 = 1$
    - $x - 5 = -3$
- The greatest power of $x$ is 1
  - There are no terms like $x^2$

## How do I solve linear equations?

- You need to use **operations** like adding, subtracting, multiplying and dividing to get **x** on its **own**
- Any operation you do to **one side** of the **equation** must also be done to the **other side**
- For example, to solve $2x + 1 = 9$ look at the +1 on the left
  - Undo this by subtracting 1 from **both sides** and **simplifying**

$$2x + 1 = 9$$

$$(-1) \qquad\qquad (-1)$$

$$2x + 1 - 1 = 9 - 1$$
$$2x = 8$$

- This equation is now **easier** to solve
- $2x$ is $2 \times x$ so undo this by dividing **both** sides by 2 and **simplifying**

$$2x = 8$$

$$(\div 2) \qquad\qquad (\div 2)$$

$$\frac{2x}{2} = \frac{8}{2}$$
$$x = 4$$

- The **solution** to the equation is $x = 4$
- Note that +1 was undone by subtracting 1
  - **addition** and **subtraction** are said to be **inverse** (opposite) operations
- Similarly, multiplying by 2 was undone by dividing by 2

Help me understand

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it

## Does the order of operations matter?

- Take $4x + 8 = 12$
  - It is **easier** to **first subtract** 8 from both sides

$$4x + 8 - 8 = 12 - 8$$
$$4x = 4$$

- **then divide** both sides by 4

$$\frac{4x}{4} = \frac{4}{4}$$
$$x = 1$$

- If you want to first divide by 4, a common **mistake** is to write $x + 8 = 3$
  - You **cannot** divide just the 4*x* and the 12 by 4
    - You must **also divide** the 8 by 4

$$\frac{4x}{4} + \frac{8}{4} = \frac{12}{4}$$
$$x + 2 = 3$$

- Then subtract 2 from both sides

$$x + 2 - 2 = 3 - 2$$
$$x = 1$$

## How do I solve linear equations with negative numbers?

- For example, solve the equation $2 - 3x = 10$
- Add $3x$ to both sides to get rid of the negative term on the left-hand side

$$2 - 3x = 10$$
$$(+3x) \qquad\qquad (+3x)$$
$$2 = 10 + 3x$$

- You can then rewrite the equation as $3x + 10 = 2$
  - And solve in the usual way
    - Subtract 10 from both sides
      $3x = -8$
    - Divide both sides by 3
      $x = -\frac{8}{3}$

- An **alternative** way to think about $2 - 3x = 10$ is by **reordering** the left-hand side
  - $2 - 3x$ is the same as $-3x + 2$
  - Reordering like this does **not change** the right-hand side
    - So the equation is $-3x + 2 = 10$

  - Some people prefer to see it like this
    - You then subtract 2 and divide by -3 as before

equation to check you got it right!

### Worked Example

Solve the equation

$$5x - 8 = 22$$

**Answer:**

Add 8 to both sides of the equation

$$5x = 22 + 8$$

Work out 22 + 8

$$5x = 30$$

Divide both sides by 5

$$x = \frac{30}{5}$$

Work out 30 ÷ 5
Keep the $x$ on the left-hand side

$$x = 6$$

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